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Lancaster et al.

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(54) **SYSTEM AND METHOD FOR PROVIDING BENEFITS BASED ON SKILL**

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See application file for complete search history.

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G07F 17/32 (2006.01)

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(57) **ABSTRACT**

Systems and methods which provide a player zero, one or more benefits based on zero, one or more skill-based inputs the player makes during one or more plays of one or more games of skill, wherein the benefits obtained are independent of any awards determined during such plays of such games of skill.

9 Claims, 6 Drawing Sheets

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FIG. 1

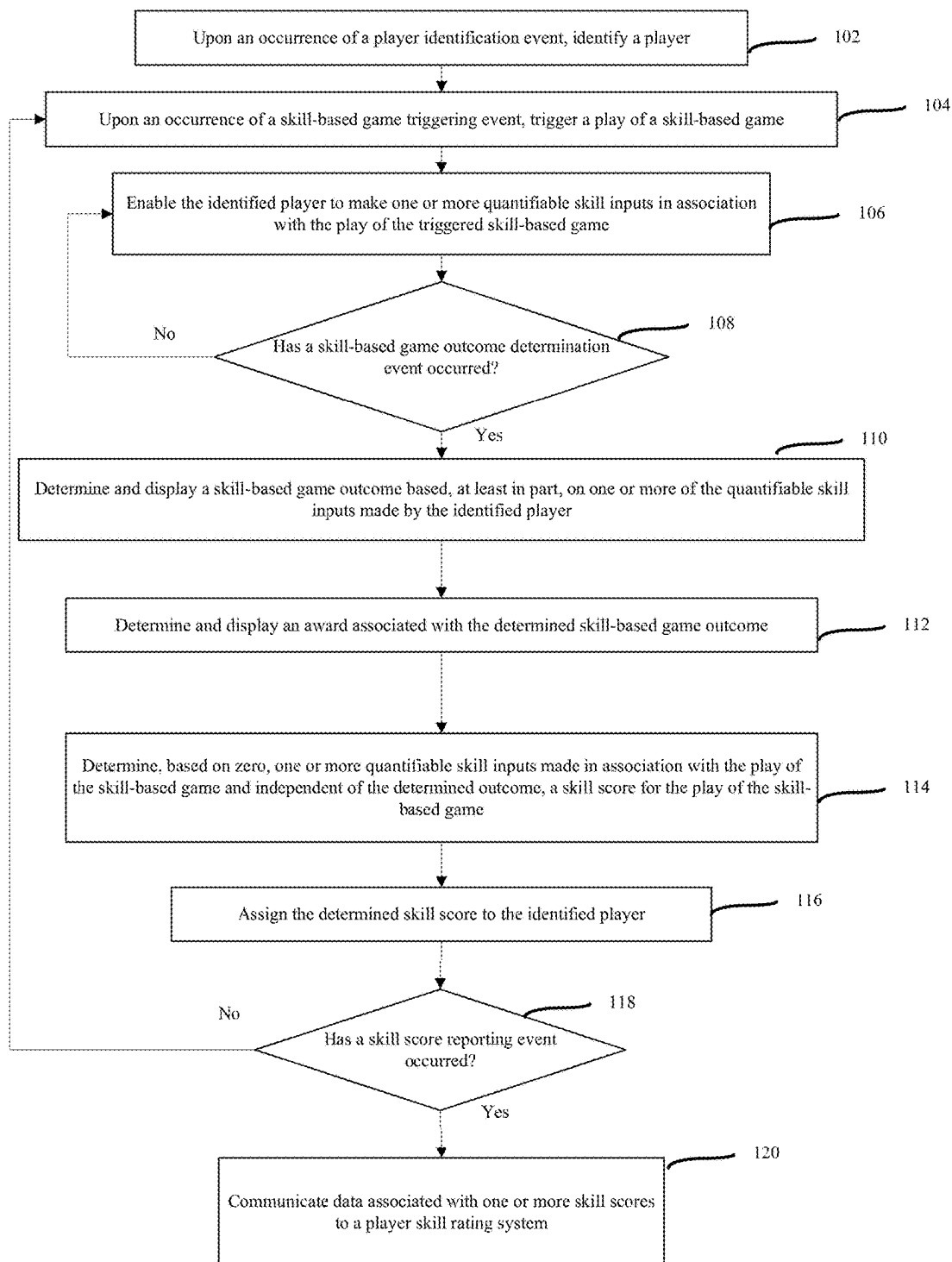


FIG. 2

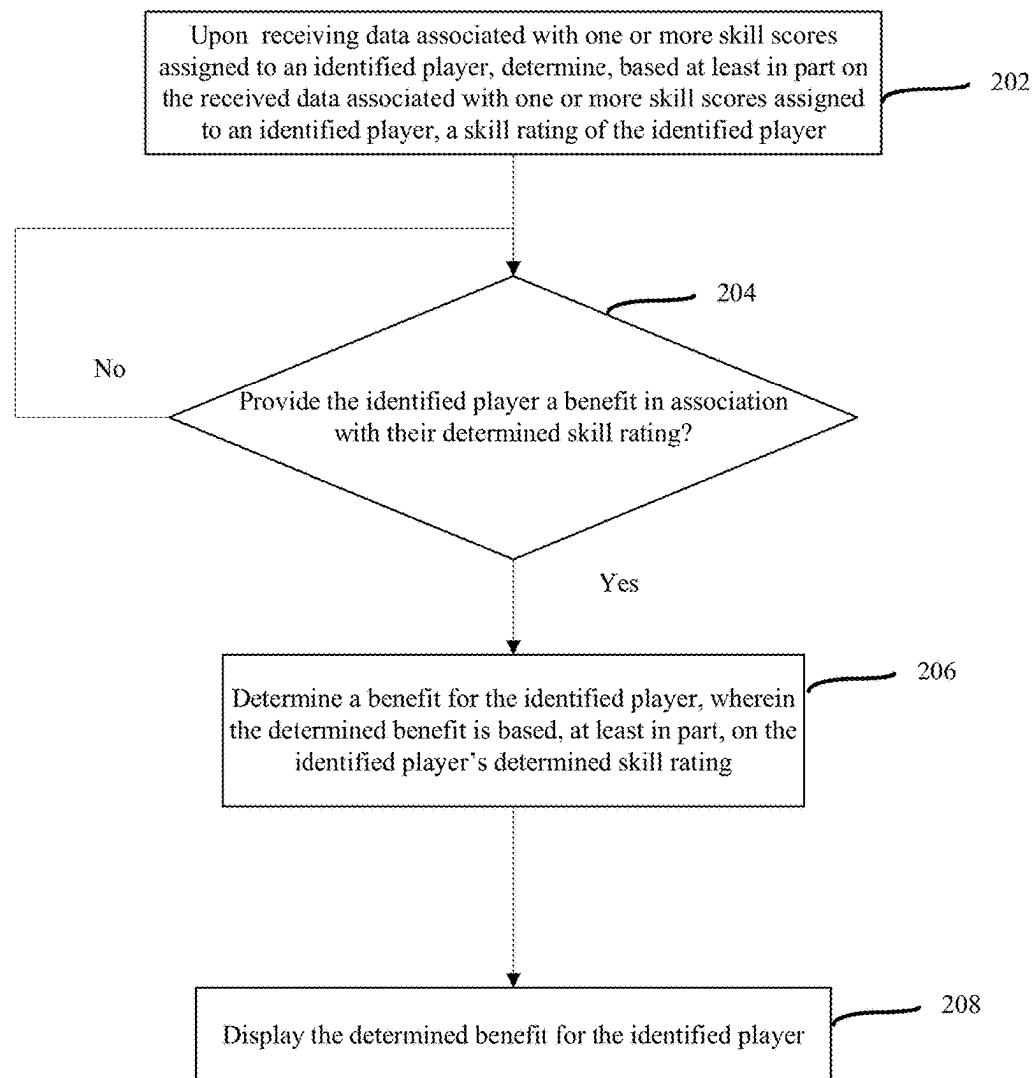


FIG. 3

1000

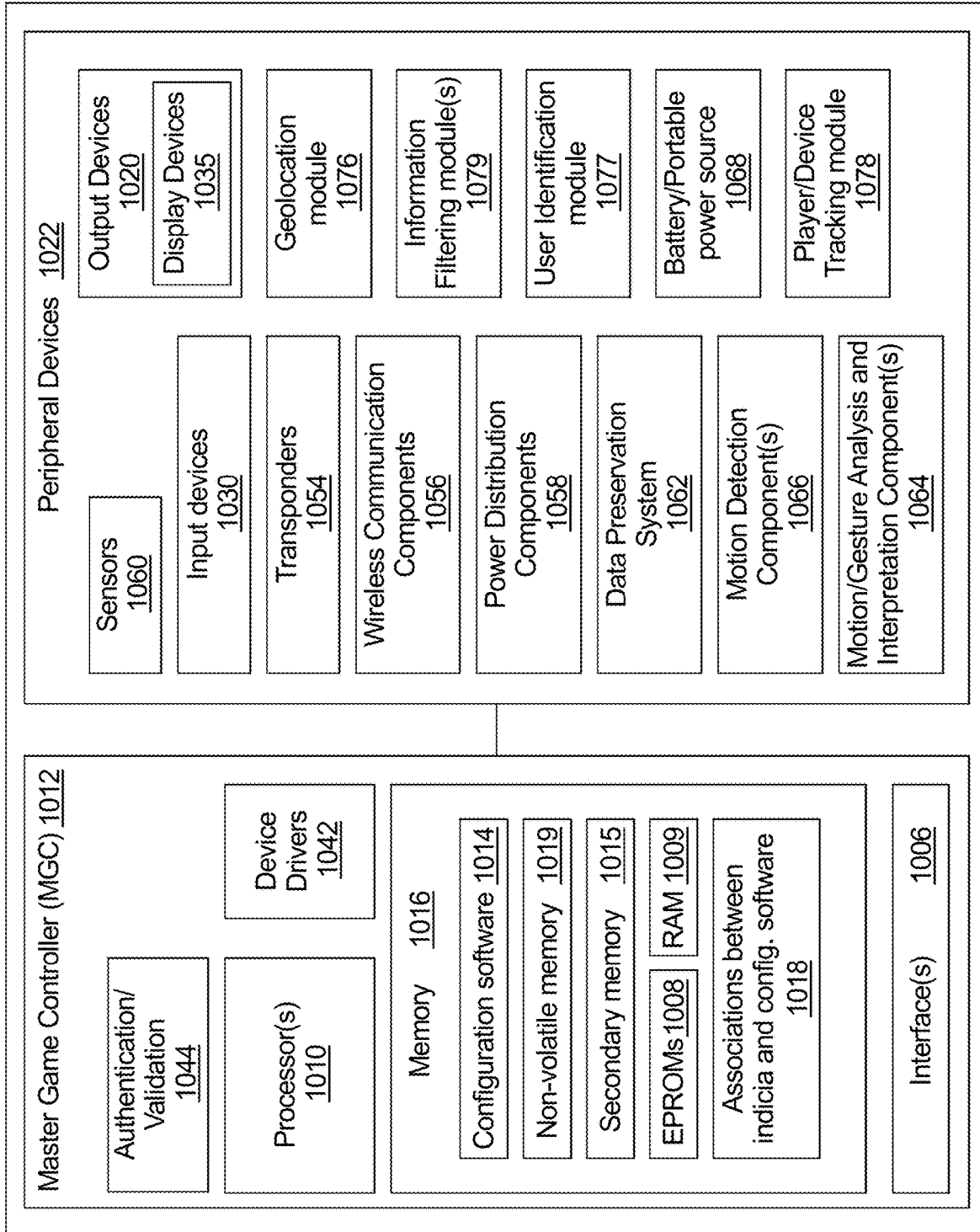


FIG. 4A

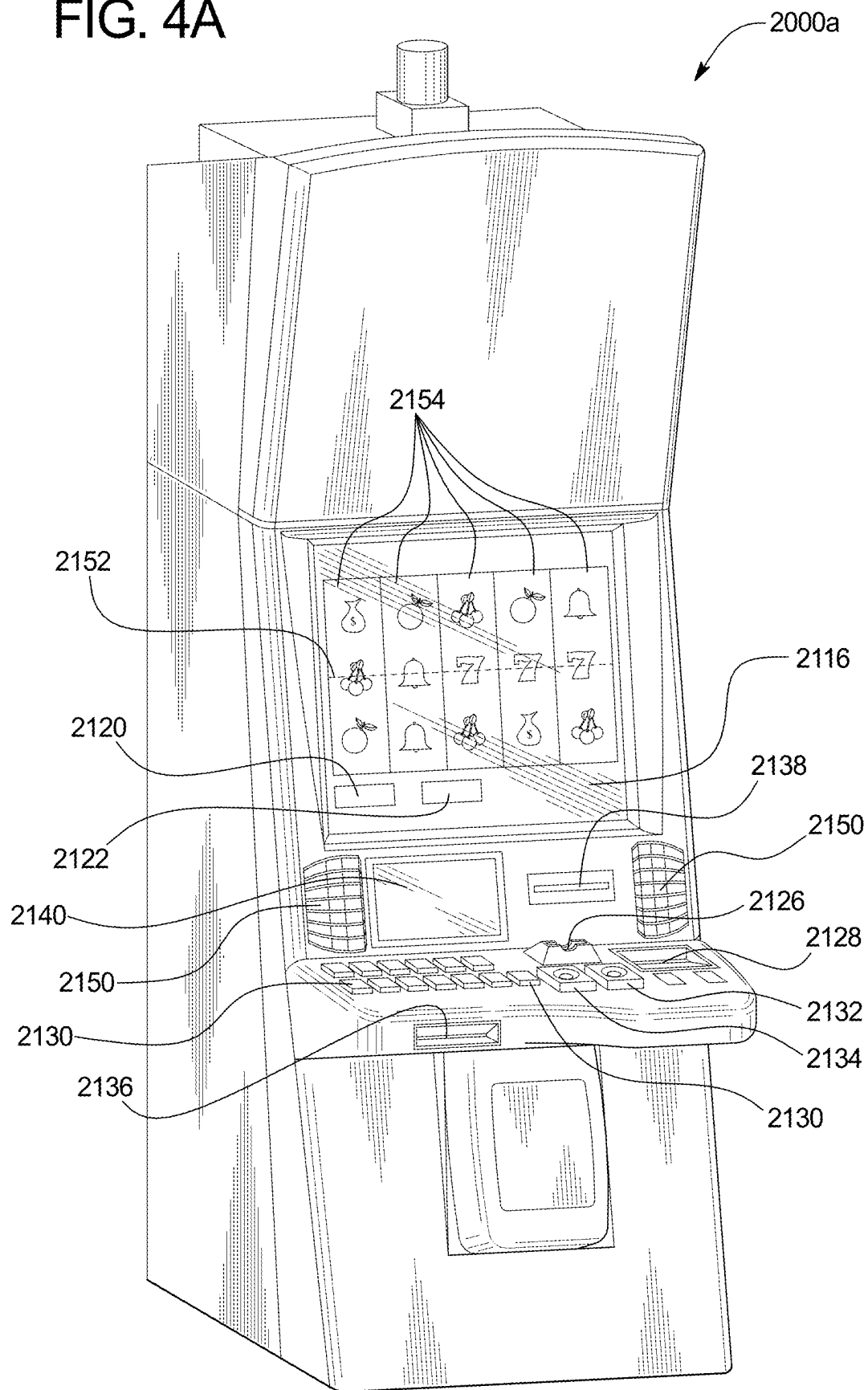


FIG. 4B

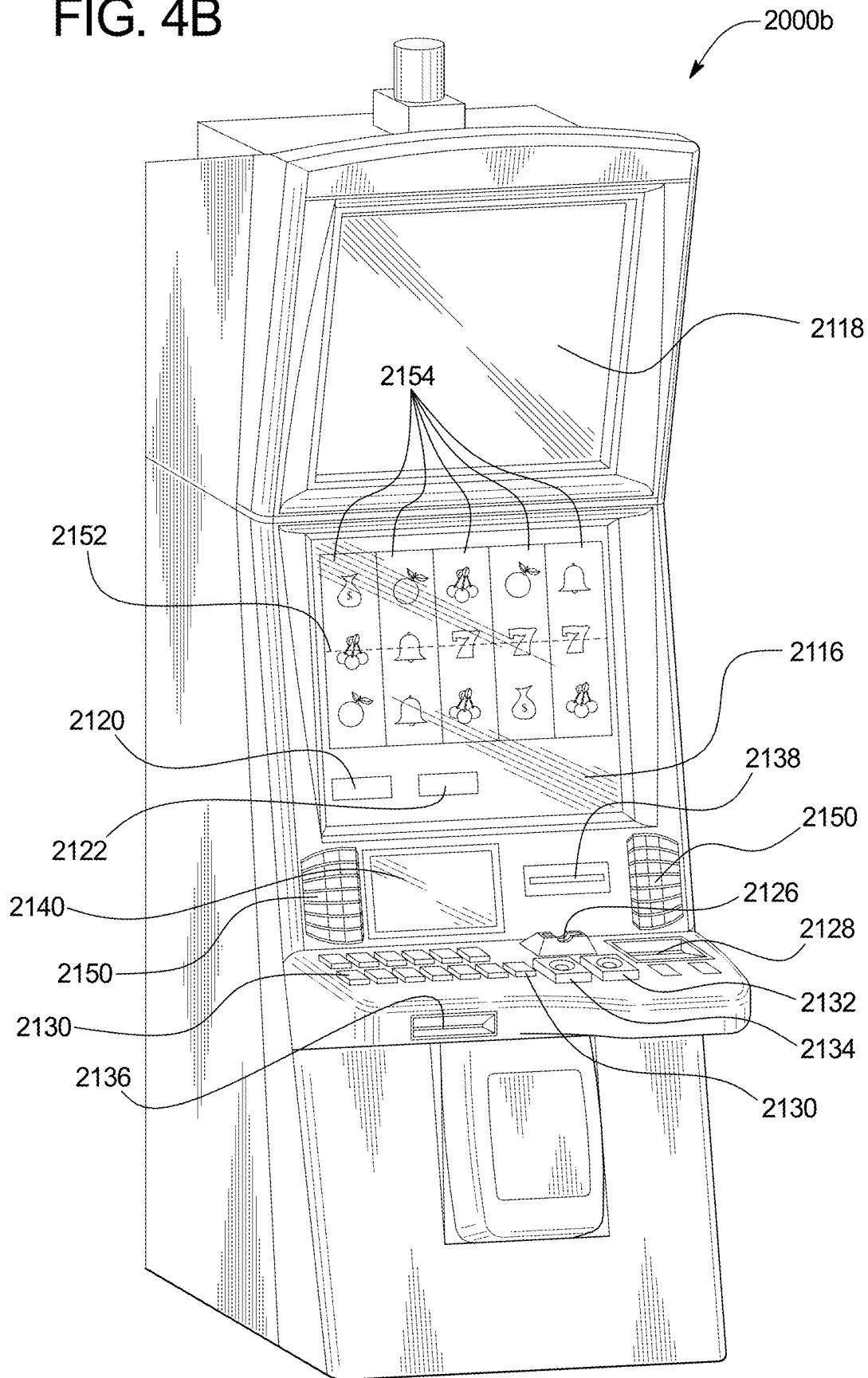
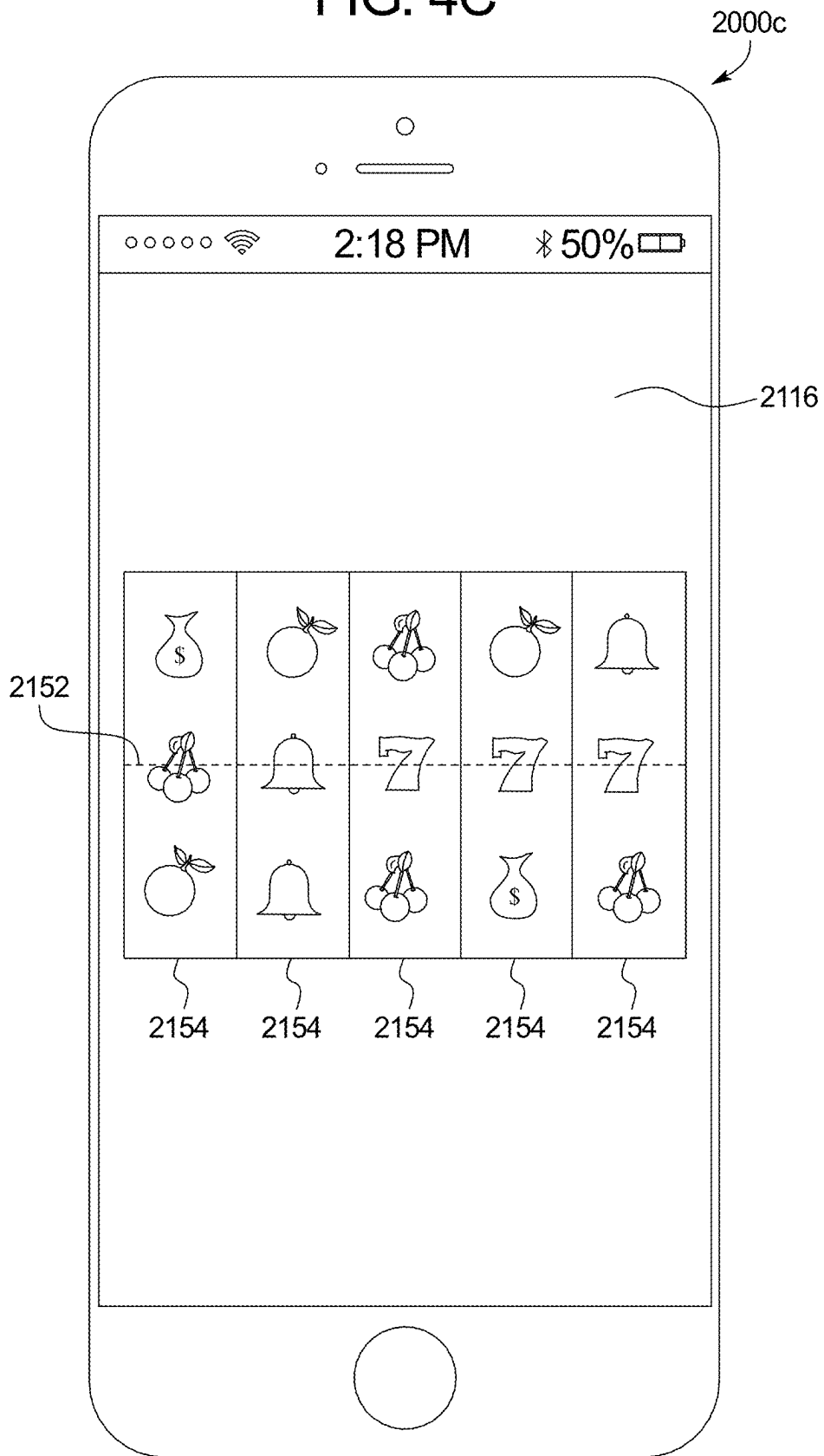


FIG. 4C



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SYSTEM AND METHOD FOR PROVIDING BENEFITS BASED ON SKILL

PRIORITY CLAIM

This application is a continuation of, claims priority to and the benefit of U.S. patent application Ser. No. 18/529,738, filed on Dec. 5, 2023, which is a continuation of, claims priority to and the benefit of U.S. patent application Ser. No. 16/983,249, filed on Aug. 3, 2020, which is a continuation of, claims priority to and the benefit of U.S. patent application Ser. No. 16/202,892, filed on Nov. 28, 2018, the entire contents of which are each incorporated by reference herein.

BACKGROUND

Gaming machines may provide players awards in primary games. Gaming machines generally require the player to place a wager to activate the primary game. The award may be based on the player obtaining a winning symbol or symbol combination and on the amount of the wager.

BRIEF SUMMARY

In certain embodiments, the present disclosure relates to an electronic gaming machine including a display device, a communication interface, an input device, a processor, and a memory device which stores a plurality of instructions. When executed by the processor, the instructions cause the processor to identify a player, and for a play of a skill-based game: receive, via the input device, a quantifiable skill input, determine, based on the received quantifiable skill input, an outcome, cause the display device to display the determined outcome, determine an award associated with the determined outcome, and cause the display device to display the determined award. When executed by the processor, the instructions cause the processor to, independent of the determined outcome, determine, based on the received quantifiable skill input, a skill score associated with the play of the skill-based game. When executed by the processor responsive to an occurrence of a skill score reporting event, the instructions cause the processor to communicate, via the communication interface, data associated with the determined skill score to a player skill rating server, wherein a player skill rating associated with the identified player is determined based on the determined skill score and a benefit is determined based on the determined player skill rating.

In certain embodiments, the present disclosure relates to a gaming system including a processor, and a memory device which stores a plurality of instructions. When executed by the processor, the instructions cause the processor to receive data associated with a determined skill score assigned to an identified player for a play of a skill-based game, the determined skill score being based on a quantifiable skill input made during the play of the skill-based game and the determined skill score being independent of any outcome determined for the play of the skill-based game. When executed by the processor, the instructions cause the processor to determine, based on the determined skill score, a player skill rating associated with the identified player. When executed by the processor responsive to an occurrence of a benefit triggering event, the instructions cause the processor to determine, based on the determined player skill rating associated with the identified player, a benefit, and cause a display, by a display device, of the determined benefit.

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In certain embodiments, the present disclosure relates to a method of operating an electronic gaming machine, the method including identifying a player, and for a play of a skill-based game: receiving, via an input device, a quantifiable skill input, determining, by a processor and based on the received quantifiable skill input, an outcome, displaying, by a display device, the determined outcome, determining, by the processor, an award associated with the determined outcome, and displaying, by the display device, the determined award. Independent of the determined outcome, the method includes determining, by the processor and based on the received quantifiable skill input, a skill score associated with the play of the skill-based game, and responsive to an occurrence of a skill score reporting event, communicating, via a communication interface, data associated with the determined skill score to a player skill rating server, wherein a player skill rating associated with the identified player is determined based on the determined skill score and a benefit is determined based on the determined player skill rating.

Additional features are described in, and will be apparent from, the following Detailed Description and the figures.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a flow chart of an example process for operating a gaming system which assigns skill scores to a player based on how that player played a game of skill compared to how that player should have played the game of skill.

FIG. 2 is a flow chart of an example process for operating a player skill rating system which determines a player skill rating based on assigned skill scores and further determines benefits in association with the player skill rating.

FIG. 3 is a schematic block diagram of one embodiment of an electronic configuration of an example gaming system disclosed herein.

FIGS. 4A and 4B are perspective views of example alternative embodiments of the gaming system disclosed herein.

FIG. 4C is a front view of an example personal gaming device of the gaming system disclosed herein.

DETAILED DESCRIPTION

Skill Rating Benefits

In various embodiments, the present disclosure relates generally to systems and methods which provide a player zero, one or more benefits based on zero, one or more skill-based inputs the player makes during one or more plays of one or more games of skill, wherein the benefits obtained are independent of any awards determined during such plays of such games of skill.

In certain embodiments, the system disclosed herein enables an identified player to play a game of skill, wherein for the play of the game of skill, the identified player makes zero, one or more quantifiable skill inputs which measure one or more aspects of the player's skill. Such player skill includes, but is not limited to: (i) physical skill, such as, but not limited to: timing, aim, physical strength or any combination thereof which is quantifiable by zero, one or more inputs made by the player in association with the game of skill; and (ii) mental skill (i.e., knowledge, reasoning, and/or strategy) which is quantifiable by one or more inputs made by the player (or the lack of any inputs made by the player) in association with the game of skill. Following these quantifiable skill inputs, the system determines and displays

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an outcome and associated award for the play of the game of skill, wherein the determined outcome is based, at least in part, on such quantifiable skill inputs.

In addition to determining and displaying an award for the play of the game of skill, the system determines a skill score for the quantifiable skill inputs, wherein the skill score is based on one or more of the quantifiable skill inputs made compared to one or more designated skill inputs, such as one or more optimal skill inputs which should have been made. In these embodiments, since a play of a game of skill is associated with an optimal way to play including zero, one or more quantifiable skill inputs which maximize the average expected award for that play of the game of skill, the system disclosed herein grades or otherwise evaluates the play of the game of skill by comparing the player's actual skill inputs to the optimal skill inputs associated with the optimal way to play the game of skill. Put differently, the system tracks a player's decisions made in a play of a game of skill and assigns a skill score to the player based on the decisions made (and not on the result of such decisions). For example, for a play of a poker game at an electronic gaming machine ("EGM"), the EGM tracks which cards of a hand the player held and based on a comparison of the cards the player held (i.e., the quantifiable skill inputs made) to which cards the player should have held if that particular hand were played optimally (i.e., the optimal quantifiable skill inputs the player should have made), the EGM determines a skill score for the player for that play of the poker game.

It should be appreciated that in tracking the quantifiable skill inputs an identified player made and assigning a skill score based on the skill inputs made compared to optimal skill inputs the player should have made, the assignment of the skill score for an individual play of a game of skill is independent of the determined outcome of that play of the game of skill. That is, to account for any random determinations which result in an award for a sub-optimally played game (i.e., a game in which a player makes one or more quantifiable skill inputs which differ from the optimal skill inputs) meeting or exceeding an award for an optimally played game, in determining a skill score for a play of a game of skill, the system disregards the actual award determined in association with the play of that game of skill. For example, two players at two EGMs are each provided the same poker hand of four cards to a Royal Flush, wherein a first player follows the optimal strategy and holds the four cards to a Royal Flush, while a second player follows a sub-optimal strategy and does not hold each of the four cards to the Royal Flush. In this example, regardless of the first player not achieving a Royal Flush outcome (and rather achieving a losing game outcome) after a draw for any replacement cards, the first EGM assigns the first player a skill score of 100 out of 100 based on the first player's skill-based inputs each being optimal skill-based inputs for that individual play of the poker game. Moreover, in this example, regardless of the second player achieving a winning game outcome after a draw for any replacement cards, the second EGM and assigns the second player a skill score of 50 out of 100 based on the second player's skill-based inputs not being optimal skill-based inputs for that individual play of the poker game. As illustrated by this example, despite the second player winning a higher award (than the first player), based on how the second player played the poker hand (relative to the first player), the second player achieved a lower skill score.

In certain embodiments, in addition to assigning a skill score for one or more plays of one or more game of skills, the EGM associated with the plays of the game of skill by

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the identified player periodically communicates data associated with the assigned skill scores to a player skill rating system. That is, at designated intervals, such as upon the end of a gaming session at an EGM, the EGM reports data regarding the player's decisions to a player skill rating system. In these embodiments, the reported data includes, but is not limited to, each of the player's decisions for each game of skill played at that EGM, each of the individual scores determined for each of the plays of the game of skill played at that EGM, or an average skill score for that gaming session which occurred at that EGM.

Following the receipt of the data associated with the assigned skill scores, the player skill rating system, such as a player skill rating server, determines a skill rating (or modifies an existing skill rating) for the identified player. In different embodiments, the skill rating accounts for one or more additional factors, such as the volume of game of skills played, the player's status and/or the player's wagering activity associated with the plays of such game of skill.

In various embodiments, in addition to maintaining a skill rating for different players, the player skill rating system periodically determines zero, one or more benefits for one or more players, wherein such benefits are based on that player's skill rating. That is, in addition to or alternatively from any benefits obtained by a player in association with a player tracking system (which provides benefits based on, amongst other elements, volume of play), the system disclosed herein provides benefits based on how skillfully the player played. As such, the player skill rating system disclosed herein enables gaming establishment operators to identify which players are relatively highly skilled players, which players are not relatively highly skilled players and then decide a benefit (i.e., an amount of reinvestment) dedicated to such different groups of players.

In addition to determining one or more benefits for one or more individual players (based on that player's individual skill rating), the player skill rating system disclosed herein enables gaming establishment operators to modify (or alternatively automatically modifies without gaming establishment operator involvement) the amount of reinvestments allocated to individual players based on demonstrated changes in that player's level of skill. For example, while a gaming establishment operator determines a first range of benefits (i.e., allocates a first level of reinvestment resources) for a player with a first, relatively low level of skill, as that player's skill level increases and that player wins more from the plays of the games of skill, the gaming establishment operator determines (or the system automatically determines) a second, lower range of benefits (i.e., allocates a second, lower level reinvestment resources) to that player. In this example, by allocated less reinvestment resources to a player that has changed from a relatively low-skilled player (and thus is, in theory, winning greater awards from the plays of the games of skill), the player skill rating system is able to reallocate such reinvestment resources to other relatively low-skilled players to subsidize these other players as such players become more familiar with the play of the games of skill.

Moreover, in addition to determining benefits for individual players based on such players determined skill ratings, the player skill rating system disclosed herein analyzes the data associated with the assigned skill scores received from the different EGMs to provide gaming establishment operators additional visibility into which types of players (e.g., relatively highly skilled players) are playing which types of EGMs. With this additional insight into how the EGMs on a gaming establishment floor are being played and

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by which types of players, the system disclosed herein provides a further tool by which gaming establishment operators may reconfigure the placement of one or more EGMs on a gaming establishment floor to maximize player's experiences at such gaming establishments.

While certain embodiments described below are directed to a primary game of skill, such as a primary skill-based wagering game which assigns a skill score based on zero, one or more skill inputs made by the player, it should be appreciated that such embodiments may additionally or alternatively be employed in association with a secondary game of skill, such as a bonus skill-based game which assigns a skill score based on zero, one or more skill inputs made by the player. Additionally, while the player's credit balance, the player's wager, and any awards are displayed as an amount of monetary credits or currency in certain of the embodiments described below, one or more of such player's credit balance, such player's wager, and any awards provided to such a player may be for non-monetary credits, promotional credits, and/or player tracking points or credits. Furthermore, the term "EGM" is used herein to refer to any suitable electronic gaming machine which enables a player to play, amongst any other games, a game of skill (or a game of partial skill), wherein the EGM comprises, but is not limited to: a slot machine, a video poker machine, a video lottery terminal, a terminal associated with an electronic table game, a video keno machine, a video bingo machine located on a casino floor, a sports betting terminal, or a kiosk, such as a sports betting kiosk.

FIG. 1 is a flowchart of an example process or method of operating the gaming system of the present disclosure. In various embodiments, the process is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the process is described with reference to the flowchart shown in FIG. 1, many other processes of performing the acts associated with this illustrated process may be employed. For example, the order of certain of the illustrated blocks or diamonds may be changed, certain of the illustrated blocks or diamonds may be optional, or certain of the illustrated blocks or diamonds may not be employed.

In various embodiments, upon an occurrence of a player identification event, as indicated by block 102 of FIG. 1, the system identifies a player. In certain embodiments wherein a player skill rating system includes or is otherwise in communication with a player tracking system server, the player identification occurs in association with a player logging into a player tracking system from an EGM. In certain embodiments wherein the player skill rating system is separate from and not in communication with a player tracking system server, the player identification occurs in association with a player logging into a player rating system. In these embodiments, the player logs into the skill rating system via one or more of: the player inserting a skill rating identification card (that has an encoded player identification number that uniquely identifies the player) into a card reader of an EGM; an establishment of a wireless communication link between an EGM and a mobile device executing an application associated with an identified player; the utilization of any suitable biometric technology or ticket technology to identify a player associated with a gaming session occurring at an EGM.

In various embodiments, upon an occurrence of a skill-based game triggering event, as indicated by block 104 of FIG. 1, the gaming system, such as an EGM, initiates or triggers a play of a skill-based game. In one embodiment, a skill-based game (or a partial skill-based game) is a primary

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game wherein a skill-based game triggering event occurs upon a player placing a wager to play the skill-based game. In another embodiment, a skill-based game (or a partial skill-based game) is a secondary or bonus game wherein a skill-based game triggering event occurs based on a displayed event associated with a wagered on play of a primary game. In another embodiment wherein the skill-based game (or a partial skill-based game) is a secondary or bonus game, a skill-based game triggering event occurs based on an event independent of any displayed event associated with a wagered on play of a primary game.

In various embodiments, the triggering skill-based game includes any suitable game of skill which utilizes zero, one or more skill inputs to determine part or all of one or more results of the game of skill. In these embodiments, such games of skill include, but are not limited to: a play of any suitable slot game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable wheel game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable card game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable offer and acceptance game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable award ladder game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable puzzle-type game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable persistence game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable selection game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable cascading symbols game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable ways to win game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable scatter pay game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable coin-pusher game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable elimination game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable stacked wilds game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable trail game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable bingo game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable video scratch-off game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable pick-until-complete game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable shooting simulation game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable racing game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable promotional game which determines part or all of the result of the play of the game based on zero, one or more

skill inputs; a play of any suitable high-low game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable lottery game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable number selection game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable dice game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable auction game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable reverse-auction game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable group game which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable game played in a service window which determines part or all of the result of the play of the game based on zero, one or more skill inputs; a play of any suitable game played on a mobile device which determines part or all of the result of the play of the game based on zero, one or more skill inputs; and/or a play of any suitable game disclosed herein which determines part or all of the result of the play of the game based on zero, one or more skill inputs.

In one embodiment, as indicated by block **106** of FIG. **1**, after an initiation of the triggered skill-based game, the gaming system, such as an EGM, enables the identified player to make one or more quantifiable skill inputs in association with the play of the triggered skill-based game. A player's skill is determined and quantified by one or more inputs (or the lack of any inputs) by the player. These determined and quantified inputs tend to measure one or more aspects of the player's skill. It should be appreciated that for purposes of this application, skill includes: (i) physical skill, such as, but not limited to: timing, aim, physical strength or any combination thereof which is quantifiable by zero, one or more inputs made by the player in association with the skill-based game; (ii) mental skill (i.e., knowledge, reasoning, and/or strategy) which is quantifiable by one or more inputs made by the player (or the lack of any inputs made by the player) in association with the skill-based game; and (iii) any other type of skill which is quantifiable by one or more inputs made by the player (or the lack of any inputs made by the player) in association with the skill-based game.

In various embodiments, the player utilizes one or more skill input devices to make one or more quantifiable skill inputs. Examples of skill input devices include, but are not limited to: joysticks, buttons, a mouse or a plurality of mice, one or more trackballs, one or more pointing devices, one or more bodily motion trackers such as motion sensing devices for human-computer interaction, touchpads, touchscreens, one or more controllers with: (1) one or more motion sensing devices, (2) one or more proximity sensing devices, (3) one or more force sensing devices (transducers), (4) one or more accelerometers, or any other suitable skill input devices.

By making one or more quantifiable skill inputs, the player manipulates, influences or otherwise controls one or more aspects of the skill-based game (and thus influences or otherwise affects the outcome of the skill-based game). In certain embodiments, different quantifiable skill inputs by the player influence a different event or a different sequence of events which occur in association with the play of the skill-based game. That is, a first quantifiable skill input (or type of quantifiable skill input) by the player results in a first

outcome, a first series of outcomes, a first event or a first sequence of events, while a second different quantifiable skill input (or type of quantifiable skill input) by the player results in a second outcome, a second series of outcomes, a second event or a second sequence of events.

In certain embodiment, the gaming system determines if a skill-based game outcome determination event has occurred as indicated by decision diamond **108**.

In one embodiment, a skill-based game outcome determination event occurs based on a displayed event associated with the play of the skill-based game. In another embodiment the skill-based game outcome determination event occurs independent of any displayed event associated with any play of any game. In another embodiments, a skill-based game outcome determination event occurs based on the actions of one or more other players. For example, if a player is playing or otherwise participating in a play of a live blackjack game from a gaming terminal associated with a remote gaming table and both the dealer and each of the players playing the game have each taken their turns for the play of the poker game, the gaming terminal determines that a skill-based game outcome determination event has occurred.

In another embodiment, a skill-based game outcome determination event occurs based on time. For example, the gaming system determines a designated amount of time for the player to play (or otherwise complete) a skill-based game and a skill-based game outcome determination event occurs when the designated amount of time elapses (or otherwise expires).

In another embodiment, the gaming system determines a designated number of quantifiable skill inputs and a skill-based game outcome determination event occurs when a player utilizes the designated number of quantifiable skill inputs. For example, the gaming system determines that a player is enabled to make ten quantifiable skill inputs to navigate a car through a maze of city streets to collect collection units. When the player makes the tenth quantifiable skill input to navigate the car through the maze of city streets, the skill-based game outcome determination event occurs.

In one embodiment, if a skill-based game outcome determination event has not occurred, the gaming system returns to block **106** and continues enabling the player to make one or more quantifiable skill inputs.

On the other hand, as indicated by block **110**, if a skill-based game outcome determination event has occurred in association with the play of the skill-based game, the gaming system determines and displays a skill-based game outcome based, at least in part, on one or more of the quantifiable skill inputs made by the identified player. After determining the outcome for the skill-based game, the gaming system determines and displays an award associated with the determined skill-based game outcome as indicated by block **112**.

In addition to determining an award and outcome for the play of the skill-based game, as indicated in block **114**, the gaming system determines, based on zero, one or more quantifiable skill inputs made in association with the play of the skill-based game and independent of the determined outcome, a skill score for the play of the skill-based game.

In certain embodiments, the play of the skill-based game is associated with zero, one or more optimal skill inputs which represent the skill inputs that a player should make during the play of the skill-based game to maximize the average expected payback percentage of the play of the skill-based game. In other words, each play of a skill-based

game is associated with one or more optimal skill inputs which are the correct skill inputs that a player should make (or should not make) when playing the skill-based game optimally (i.e., playing the skill-based game which employs a strategy associated with the highest average expected payback percentage of the various strategies available to be employed). In these embodiments, the gaming system determines a skill score based on one or more of the quantifiable skill inputs made compared to one or more of such optimal skill inputs. That is, the system grades or otherwise evaluates the play of the skill-based game by comparing the player's actual skill inputs to the optimal skill inputs associated with the optimal way to play the skill-based game.

For example, for an initial poker hand of a jack of hearts, a queen of spades, a ten of hearts, a four of hearts, and a king of hearts, based on the payouts available for this initial hand (when accounting for the paytable employed and the remaining cards available to be dealt with the draw) and the probability of obtaining those payouts for this poker hand, the gaming system determines that: (i) holding the jack of hearts, ten of hearts and king of hearts to play for obtaining a royal flush has the highest expected value; (ii) holding the jack of hearts, the ten of hearts, the four of hearts, and the king of hearts to play for obtaining a flush has a lower expected value; and (iii) holding the jack of hearts, the queen of spades, the ten of hearts, and the king of hearts to play for obtaining a straight has an even lower expected value. In this example, if the player was initially dealt the poker hand of the jack of hearts, the queen of spades, the ten of hearts, the four of hearts, and the king of hearts, and the player decided to play the poker game optimally and held the jack of hearts, ten of hearts and king of hearts (i.e., the player's skill inputs for the play of the poker game) to play for obtaining a royal flush, then regardless of whether or not the player obtained the royal flush, the gaming system assigns a skill score of 100 out of 100 to the player for that play of the poker game. On the other hand, if the player decided to play the poker game sub-optimally and held the jack of hearts, the ten of hearts, the four of hearts, and the king of hearts (i.e., the player's skill inputs for the play of the poker game) to play for obtaining a flush, then regardless of whether or not the player obtained the flush, the gaming system assigns a skill score of 95 out of 100 to the player for that play of the poker game. Additionally, in this example, if the player decided to play the poker game sub-optimally and held the jack of hearts, the queen of spades, the ten of hearts, and the king of hearts (i.e., the player's skill inputs for the play of the poker game) to play for obtaining a straight, then regardless of whether or not the player obtained the straight, the gaming system assigns a skill score of 65 out of 100 to the player for that play of the poker game. As illustrated by this example, the system tracks a player's decisions made in a play of a game of skill and assigns a skill score to the player based on the decisions made (relative to optimal decisions the player should have made).

It should be appreciated that in tracking the quantifiable skill inputs an identified player made and assigning a skill score based on the skill inputs made compared to optimal skill inputs the player should have made, as indicated above, the assignment of the skill score for an individual play of a skill-based game is independent of the determined outcome of that play of the skill-based game. That is, to account for zero, one or more random determinations which result in an award for a sub-optimally played game (i.e., a game in which a player makes one or more quantifiable skill inputs which differ from the optimal skill inputs) meeting or exceeding an award for an optimally played game, in

determining a skill score for a play of a skill-based game, the system disregards the actual award determined in association with the play of that skill-based game.

Building on the above example, if two EGMs initially deal two players the same poker hand of a jack of hearts, a queen of spades, a ten of hearts, a four of hearts, and a king of hearts, both players made the same skill inputs to hold the jack of hearts, ten of hearts and king of hearts (or inputs to discard the four of hearts and queen of spades), and following the draw of replacement cards, one player obtained a royal flush poker hand and the other player did not obtain a royal flush poker hand, the two EGMs respectively assign each of the two players the same skill score of 100 out of 100 for their optimal play of the poker game (regardless of the actual outcome of the play of the optimal play of the poker game).

In another example, if: (a) a first EGM initially deals to a first player a poker hand of a jack of hearts, a queen of spades, a ten of hearts, a four of hearts, and a king of hearts, (b) the first player made skill inputs to hold the jack of hearts, ten of hearts and king of hearts (or inputs to discard the four of hearts and queen of spades), and (c) the poker draw resulted in a losing hand associated with an award of zero credits, then the EGM assigns a skill score of 100 out of 100 to the first player for that optimal play of the poker game. In this example, if a second EGM initially deals to a second player the same poker hand of a jack of hearts, a queen of spades, a ten of hearts, a four of hearts, and a king of hearts, (b) the second player made skill inputs to hold the jack of hearts, the ten of hearts, the four of hearts, and the king of hearts (or inputs to discard the queen of spades) to play for obtaining a flush, and (c) the poker draw resulted in a winning flush poker hand associated with an award of thirty credits, then the EGM assigns a skill score of 65 out of 100 to the second player for the sub-optimal play of the poker game. As illustrated by this example, despite the second player obtaining a higher award than the first player, since the first player played the initially dealt poker hand optimally (and the second player did not), the first player is assigned a higher skill score for this particular play of the poker game.

Following the determination of a skill score for the play of the skill-based game, the gaming system assigns the determined skill score to the identified player as indicated in block 116 of FIG. 1. That is, since the system disclosed herein provides one or more benefits to an identified player based on one or more skill inputs made which are quantified by a skill score, the system assigns such a skill score to the identified player and retains the assigned skill score in association with the identified player's gaming session.

Following the assignment of a skill score to the identified player in association with one or more plays of the skill-based game, the system determines whether a skill score reporting event has occurred as indicated in diamond 118 of FIG. 1.

In certain embodiments, a skill score reporting event occurs following the conclusion of a play of a skill-based game. In certain embodiments, a skill score reporting event occurs following the conclusion of a designated quantity of plays of one or more skill-based games. In certain embodiments, a skill score reporting event occurs following the conclusion of a gaming session, such as upon a player logging out of an EGM. In certain embodiments, a skill score reporting event automatically occurs periodically, such as upon an elapsed period of time.

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In these embodiments, if the skill score reporting event has not occurred, the gaming system returns to block **104** and awaits for another occurrence of a skill-based game triggering event.

On the other hand, upon the occurrence of the skill score reporting event, the gaming system communicates data associated with one or more skill scores to a player skill rating system as indicated by block **120**. In one embodiment, the data associated with one or more skill scores includes the determined skill scores. In another embodiment, the data associated with one or more skill scores includes the determined skill scores and one or more of the skill inputs associated with obtaining the determined skill scores. In another embodiment, the data associated with one or more skill scores includes the determined skill scores for each of the plays of the skill-based game. In another embodiment, the data associated with one or more skill scores includes a sampling of a designated quantity of the determined skill scores from a sampling of a designated quantity of plays of the skill-based game. In another embodiment, the data associated with one or more skill scores includes an average determined skill score from a plurality of plays of the skill-based game. In another embodiment, the data associated with one or more skill scores also includes the outcomes and/or associated awards resulting from one or more plays of one or more skill-based game. In another embodiment, the data associated with one or more skill scores includes an average determined skill score from a plurality of plays of the skill-based game. In another embodiment, the data associated with one or more skill scores includes the one or more skill inputs made and the optimal skill inputs the player should have made such that the player skill rating system additionally or alternatively determines the skill scores.

In certain embodiments, the player skill rating system is part of a player tracking system wherein the gaming system, such as the EGM which determined the skill scores for one or more plays of one or more skill-based games, communicates the data associated with the one or more skill scores to a player tracking system server. In certain other embodiments, the player skill rating system is independent of a player tracking system server wherein the gaming system, such as the gaming terminal associated with a remote table game which determined the skill scores for one or more plays of one or more skill-based games, communicates the data associated with the one or more skill scores to a player skill rating server. In these embodiments, the player skill rating system then utilizes at least the reported skill score data to provide zero, one or more benefits to the player of the skill-based game.

FIG. 2 is a flowchart of an example process or method of operating the player skill rating system of the present disclosure. In various embodiments, the process is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the process is described with reference to the flowchart shown in FIG. 2, many other processes of performing the acts associated with this illustrated process may be employed. For example, the order of certain of the illustrated blocks or diamonds may be changed, certain of the illustrated blocks or diamonds may be optional, or certain of the illustrated blocks or diamonds may not be employed.

In certain embodiments, upon the player skill rating system receiving data associated with one or more skill scores assigned to an identified player, as indicated in block **202**, the player skill rating system determines, based at least

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in part on the received data associated with one or more skill scores assigned to an identified player, a skill rating of the identified player.

In certain embodiments wherein the identified player does not have a previously determined skill rating, the player skill rating system determines a skill rating for the identified player based on the received data associated with one or more skill scores assigned to an identified player. In these embodiments, the established skill rating for the player is based on the skill scores obtained in association with the gaming session which occurred at the EGM.

In certain embodiments wherein the identified player has a previously determined skill rating, the player skill rating system determines a modified skill rating for the identified player (which may be the same or different from the previously determined skill rating) based on the previously determined skill rating and the received data associated with one or more skill scores assigned to an identified player. In these embodiments, the skill rating for the player is based on the skill scores obtained in association with the gaming session which occurred at the EGM as well as one or more previously determined skill scores obtained in association with one or more gaming sessions which occurred at one or more EGMs.

In one embodiment, the determined skill rating (or modified skill rating) for the identified player is additionally based on the wagering activity of the identified player. In another embodiment, the determined skill rating (or modified skill rating) for the identified player is additionally based on the gameplay activity of the identified player. In another embodiment, the determined skill rating (or modified skill rating) for the identified player is additionally based on the player tracking status of the identified player.

Following the determination (or modification) of a skill rating for an identified player, the player skill rating system determines whether to provide the identified player a benefit in association with their determined skill rating as indicated in diamond **204**. That is, the player skill rating system determines whether or not a benefit triggering event has occurred for an identified player.

In certain embodiments, a benefit triggering event occurs based on a displayed event associated with the play of the skill-based game. In certain embodiments, a benefit triggering event occurs independent of any displayed event associated with any play of any game. In certain embodiments, a benefit triggering event occurs based on an elapsed period of time. In certain embodiments, a benefit triggering event occurs based on the satisfaction of one or more conditions.

If the player skill rating system determines not to provide the identified player a benefit at that time, the player skill rating system returns to diamond **204** and awaits for another determination of whether or not to provide a benefit. It should be appreciated that while awaiting for an occurrence of a benefit triggering event for an identified player, data associated with zero, one or more skill scores may be received from zero, one or more EGMs resulting in the modification of the skill rating of the identified player. As such, the player skill rating system disclosed herein dynamically rates (and rerates) players based on the decisions such players are making in association with one or more skill-based games played at one or more EGMs associated with the player skill rating system.

On the other hand, if the player skill rating system determines to provide the identified player a benefit, as indicated in blocks **206** and **208**, the player skill rating system determines and causes a display of a benefit for the identified player, wherein the determined benefit is based, at

least in part, on the identified player's determined skill rating. As such, in addition to or alternatively from any benefits obtained by a player in association with a player tracking system (which provides benefits based on, amongst other elements, volume of play), the system disclosed herein provides benefits based on how skillfully the player played. Accordingly, the player skill rating system enables gaming establishment operators to identify which players are relatively highly skilled players, which players are not relatively highly skilled players and then decide (or manually or automatically) a benefit (i.e., an amount of reinvestment) dedicated to such different groups of players.

In certain embodiments, the player skill rating system automatically provides the determined benefit to the identified player. In these embodiments, the player skill rating system notifies the player of the determined benefit in any suitable manner, such as via a display device of an EGM where the player is currently playing at, via a notification displayed by a gaming establishment application being executed on the player's mobile device, via an e-mail, via one or more SMS or text messages, via one or more social media postings or via any other suitable medium. In certain embodiments, the player skill rating system displays the determined benefit to a gaming establishment operator, such as via a gaming establishment operator workstation, and then enables the gaming establishment operator to accept, reject or modify the determined benefit being provided to the identified player.

In certain embodiments, the determination of the benefit is based on the player and/or an identity of the player. In one such embodiment, different players having different status with the gaming establishment are offered different benefits. In another such embodiment, the system determines the benefit based on the historical preferences of the player. In another such embodiment, the system determines the benefit based on the player's wagering activity. In another such embodiment, the system determines the benefit based on the player's gaming establishment retail purchasing activity.

In certain embodiments, the determination of the benefit is additionally or alternatively based on one or more parameters of the gaming establishment, such as when the benefit is determined, and/or an amount of wagering activity occurring at the gaming establishment.

In certain embodiments, the determined benefit includes an entry into a drawing for an award. In certain of these embodiments, the drawing includes the single selection of an entry for a single player to win an award. In certain of these embodiments, the drawing includes multiple selections of multiple entries for one or more players to win one or more awards. In one such embodiment, a quantity of entries in the drawing is based on the player skill rating, wherein different player skill ratings are associated with different quantities of drawing entries. It should be appreciated that while illustrating as offering the player entries into a drawing to win a static award amount, in various embodiments, the entries into the drawing are to win one or more of a quantity of player tracking points, one or more player tracking point multipliers, a quantity of promotional credits, an increased player tracking rank at a gaming establishment for a period of time, a quantity of credits useable in an associated social/mobile game, a non-cash award, such as an automobile, a house, and/or a vacation.

In certain embodiments, the determined benefit includes one or more free (or reduced cost) plays of one or more games of chance (or games of skill as disclosed herein). In certain other embodiments, the determined benefit includes one or more free (or reduced cost) plays of one or more

online games of chance (or games of skill as disclosed herein) accessible remote from the gaming establishment. In one such embodiment, a quantity of plays of the game of chance (or game of skill) is based on the skill rating of the player, wherein different skill ratings are associated with different quantities of plays of the game of chance. In these embodiments, such games of chance include, but are not limited to: a play of any suitable slot game; a play of any suitable wheel game; a play of any suitable card game; a play of any suitable offer and acceptance game; a play of any suitable award ladder game; a play of any suitable puzzle-type game; a play of any suitable persistence game; a play of any suitable selection game; a play of any suitable cascading symbols game; a play of any suitable ways to win game; a play of any suitable scatter pay game; a play of any suitable coin-pusher game; a play of any suitable elimination game; a play of any suitable stacked wilds game; a play of any suitable trail game; a play of any suitable bingo game; a play of any suitable video scratch-off game; a play of any suitable pick-until-complete game; a play of any suitable shooting simulation game; a play of any suitable racing game; a play of any suitable promotional game; a play of any suitable high-low game; a play of any suitable lottery game; a play of any suitable number selection game; a play of any suitable dice game; a play of any suitable skill game; a play of any suitable auction game; a play of any suitable reverse-auction game; a play of any suitable group game; a play of any suitable game in a service window; a play of any suitable game on a mobile device; and/or a play of any suitable game disclosed herein.

It should be appreciated that in these embodiments, since one or more entries into one or more drawings, and/or one or more plays of one or more games of chance (or games of skill) may or may not result in an award for the player, such benefits are opportunities to obtain one or more benefits. That is, while such benefits have an average expected value (i.e., a theoretical value which is based on the awards available and the probability of obtaining any of such awards), since one or more random determinations subsequently occur in association with the determined benefit and an actual value of the determined benefit (i.e., the benefit associated with the determined benefit) is based on the results of such random determinations (wherein the actual value may be a lucrative value or a value of zero), these determined benefits may be viewed as opportunities to obtain a benefit.

In certain embodiments, the determined benefit includes a static award, such as a quantity of credits or a prize. In one such embodiment, the amount of the static award is based on the skill rating of the player, wherein different skill ratings are associated with different static awards.

In certain embodiments, the determined benefit includes a conditional award that the player becomes eligible for based on the satisfaction of one or more criteria. For example, if a player associated with an benefit of a conditional award places a wager on a play of a game at the gaming establishment within a designated period of time, the system determines that the criteria associated with the conditional award is satisfied and the conditional award vests (i.e., the award becomes available to the player). In another example, if a player associated with an benefit of a conditional award maintains a designated amount of wagering activity for a designated period of time, the system determines that the criteria associated with the conditional award is satisfied and the conditional award vests (i.e., the award becomes available to the player). In another example, if a player associated with an benefit of a conditional award completes a desig-

nated amount of gaming establishment retail purchases within a designated period of time, the system determines that the criteria associated with the conditional award is satisfied and the conditional award vests (i.e., the award becomes available to the player). In another example, if a player takes one or more actions within a designated period of time (e.g., such as sharing a social media post regarding the gaming establishment or commenting on a social media post associated with the gaming establishment), the system determines that the criteria associated with the conditional award is satisfied and the conditional award vests (i.e., the award becomes available to the player). It should be appreciated that any suitable criteria associated with the gaming establishment, associated with an entity having a relationship with the gaming establishment or independent of the gaming establishment may be utilized to determine whether or not the player satisfies such criteria and thus becomes eligible to receive a benefit associated with the determined benefit.

In certain embodiments, the determined benefit includes a quantity of player tracking points. In certain embodiments, different player skill ratings are associated with different amounts of provided player tracking points. In certain embodiments, the determined benefit includes a player tracking point modifier applicable to a quantity of player tracking points subsequently earned by the player. In these embodiments, the system determines a benefit of a player tracking point modifier which can be activated for subsequently earned player tracking points. In certain embodiments, the determined benefit additionally or alternatively includes a player tracking point modifier applicable to a quantity of player tracking points previously earned by the player. In certain embodiments, different player skill ratings are associated with different player tracking point modifiers.

In certain embodiments, the determined benefit includes a quantity of non-cashable promotional credits usable to fund one or more plays of one or more games of chance or games of skill. In certain other embodiments, the determined benefit additionally or alternatively includes a quantity of non-cashable promotional credits usable to fund one or more plays of one or more online games of chance or games of skill accessible remote from the gaming establishment. In certain embodiments, different player skill ratings are associated with different quantities of non-cashable promotional credits. It should be appreciated that while such non-cashable promotional credits have an actual value (i.e., the average expected value of the game(s) played which are funded by such promotional credits), since the games played using such non-cashable promotional credits may result in one or more awards different than the average expected value of such games, the player may ultimately realize a different valued benefit. In certain embodiments, the determined benefit includes a quantity of virtual credits usable for a play of a social or non-wagering game. In certain embodiments, different player skill ratings are associated with different quantities of virtual credits earned as benefits and usable for a play of a social or non-wagering game.

In certain embodiments, the determined benefit includes one or more features usable in association with a play of a game. That is, the determined benefit is associated with one or more plays of one or more games, such as a modification to one or more game play features and/or an addition of one or more game play features. In certain of these embodiments, the determined benefit includes altering or otherwise modify one or more features, aspects or parameters of one or more subsequently played games. In these embodiments, a determined benefit includes any suitable feature which

modifies any aspect of any game subsequently played by the player which is associated with the offered benefit. In various embodiments, one or more features employed as a determined benefit to modify one or more aspects of one or more such games subsequently played include, but are not limited to: a feature modifying one or more symbols available to be generated for a subsequent play of a game associated with the offered benefit; a feature modifying one or more wild symbols available to be generated for a subsequent play of a game associated with the offered benefit; a feature modifying a quantity of reels to be used for a subsequent play of a game associated with the offered benefit; a feature modifying which of a plurality of reel are to be used for a subsequent play of a game associated with the offered benefit; a feature modifying a deck of playing cards to be used for a subsequent play of a game associated with the offered benefit; a feature modifying a quantity of playing cards to be used for a subsequent play of a game associated with the offered benefit; a feature modifying a quantity of poker hands to be dealt for a subsequent play of a game associated with the offered benefit; a book-end wild symbols feature; a stacked wild symbols feature; an expanding wild symbols feature; a retrigger symbol feature; an anti-terminator symbol feature; a locking reel feature, a locking symbol position feature; a modifier, such as a multiplier, feature; a feature modifying an amount of credits of a credit balance; a feature modifying an amount of promotional credits; a feature modifying a placed wager amount; a feature modifying a placed side wager amount; a feature modifying a rate of earning player tracking points; a feature modifying a rate of earning promotional credits; a feature modifying a rate of earning virtual credits; a feature modifying a number of wagered on paylines; a feature modifying a wager placed on one or more paylines (or on one or more designated paylines); a feature modifying a number of ways to win wagered on; a feature modifying a wager placed on one or more ways to win (or on one or more designated ways to win); a feature modifying an average expected payback percentage of a subsequent play of a game associated with the offered benefit; a feature modifying an average expected payout of a subsequent play of a game associated with the offered benefit; a feature modifying one or more awards available; a feature modifying a range of awards available; a feature modifying a type of awards available; a feature modifying one or more progressive awards; a feature modifying which progressive awards are available to be won; a feature modifying one or more modifiers, such as multipliers, available; a feature modifying an activation of a reel (or a designated reel); a feature modifying an activation of a plurality of reels; a feature modifying a generated outcome (or a designated generated outcome); a feature modifying a generated outcome (or a designated generated outcome) associated with an award over a designated value; a feature modifying a generated outcome (or a designated generated outcome) on a designated payline; a feature modifying a generated outcome (or a designated generated outcome) in a scatter configuration; a feature modifying a winning way to win (or a designated winning way to win); a feature modifying a designated symbol or symbol combination; a feature modifying a generation of a designated symbol or symbol combination on a designated payline; a feature modifying a generation of a designated symbol or symbol combination in a scatter configuration; a feature modifying a triggering event of a play of a secondary or bonus game; a feature modifying an activation of a secondary or bonus display (such as an award generator); a feature modifying a quantity of activations of

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a secondary or bonus display (e.g., a feature modifying a quantity of spins of an award generator); a feature modifying a quantity of sections of a secondary or bonus display (e.g., a feature modifying a quantity of sections of an award generator); a feature modifying one or more awards of a secondary or bonus display; a feature modifying an activation of a community award generator; a feature modifying a quantity of activations of a community award generator; a feature modifying a quantity of sections of a community award generator; a feature modifying one or more awards of a community award generator; a feature modifying a generated outcome (or a designated generated outcome) in a secondary game; a feature modifying a quantity of picks in a selection game; a feature modifying a quantity of offers in an offer and acceptance game; a feature modifying a quantity of moves in a trail game; a feature modifying an amount of free spins provided; a feature modifying a game terminating or ending condition; a feature modifying an availability of a secondary game; and/or a feature modifying any game play feature associated with any play of any game disclosed herein.

It should be appreciated that in these embodiments, since the conditional prize, the player tracking points, the promotional credits, the virtual credits, the player tracking point modifier and the feature(s) have static values, such benefits are associated with known benefits. That is, unlike certain benefits, such as an entry into a drawing or a play of a game of chance, which may result in the player obtaining no value or which may result in the player obtaining a positive valued benefit, other benefits, such as a quantity of player tracking points or a quantity of virtual credits will result in the player obtaining value (upon the satisfaction of zero, one or more conditions).

It should be appreciated that by dynamically determining a skill rating of a player and providing zero, one or more benefits based, at least in part, on the determined skill rating, the player skill rating system disclosed herein enables gaming establishment operators to modify (or alternatively automatically modifies without gaming establishment operator involvement) the amount of reinvestments allocated to individual players based on demonstrated changes in that player's level of skill. That is, by periodically modifying the skill rating of a particular player (based on the skill scores reported by one or more EGMs which are representative of zero, one or more player decisions in one or more plays of one or more skill-based games), the player skill rating system enables a gaming establishment operator to periodically modify the benefit level attributed to such a particular player. For example, while a gaming establishment operator determines a first range of benefits (i.e., allocates a first level of reinvestment resources) for a player with a first, relatively low level of skill, as that player's skill level increases and that player wins more from the plays of the games of skill, the gaming establishment operator determines (or the system automatically determines) a second, lower range of benefits (i.e., allocates a second, lower level reinvestment resources) to that player. In this example, by allocated less reinvestment resources to a player that has changed from a relatively low-skilled player (and thus is, in theory, winning greater awards from the plays of the games of skill), the player skill rating system is able to reallocate such reinvestment resources to other relatively low-skilled players to subsidize these other players as such players become more familiar with the play of the games of skill.

Moreover, in addition to determining benefits for individual players based on such players determined skill ratings, the player skill rating system disclosed herein analyzes

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the data associated with the assigned skill scores received from the different EGMs to provide gaming establishment operators additional visibility into which types of players (e.g., relatively highly skilled players) are playing which types of EGMs. With this additional insight into how the EGMs on a gaming establishment floor are being played and by which types of players, the system disclosed herein provides a further tool by which gaming establishment operators may reconfigure the placement of one or more EGMs on a gaming establishment floor to maximize player's experiences at such gaming establishments.

In one embodiment, the gaming system provides a group gaming aspect to the skill-based games disclosed herein. In one such embodiment, the game is a cooperative community skill-based game wherein a plurality of players cooperate or play together to win one or more awards. In another such embodiment, the games disclosed herein a competition community skill-based game wherein a plurality of players compete or player against each other to win one or more awards.

In different embodiments, one or more awards provided in association with the skill-based games disclosed herein include one or more of: a quantity of monetary credits, a quantity of non-monetary credits, a quantity of promotional credits, a quantity of player tracking points, a progressive award, a modifier, such as a multiplier, a quantity of free plays of one or more games, a quantity of plays of one or more secondary or bonus games, a multiplier of a quantity of free plays of a game, one or more lottery based awards, such as lottery or drawing tickets, a wager match for one or more plays of one or more games, an increase in the average expected payback percentage for one or more plays of one or more games, one or more comps, such as a free dinner, a free night's stay at a hotel, a high value product such as a free car, or a low value product, one or more bonus credits usable for online play, a lump sum of player tracking points or credits, a multiplier for player tracking points or credits, an increase in a membership or player tracking level, one or more coupons or promotions usable within and/or outside of the gaming establishment (e.g., a 20% off coupon for use at a convenience store), virtual goods associated with the gaming system, virtual goods not associated with the gaming system, an access code usable to unlock content on an internet.

In one embodiment, the gaming system causes at least one display device of an EGM to display the skill-based game and/or the player skill rating benefits obtained. In another embodiment, in addition or in alternative to each EGM displaying the skill-based game and/or the player skill rating benefits obtained, the gaming system causes one or more community or overhead display devices and/or personal devices to display part or all of the skill-based game and/or the player skill rating benefits obtained to one or more other players or bystanders either at a gaming establishment or viewing over a network, such as the internet. In another embodiment, in addition or in alternative to each EGM displaying the skill-based game and/or the player skill rating benefits obtained, the gaming system causes one or more internet sites to each display the skill-based game and/or the player skill rating benefits obtained such that a player is enabled to log on from a personal web browser.

Alternative Embodiments

It should be appreciated that in different embodiments, one or more of:

- i. when a skill-based game triggering event, a skill-based game outcome determination event, a skill score reporting event and/or a benefit triggering event occurs;
- ii. what type of skill-based game to initiate;
- iii. which type of skill to associate with a skill-based game;
- iv. which type of player skill inputs to enable one or more players to make;
- v. a quantity of player skill inputs to enable one or more players to make;
- vi. a quantity of players participating in a skill-based game;
- vii. one or more amounts of time allotted for a play of a skill-based game;
- viii. how to determine a skill score;
- ix. when to determine a skill score;
- x. how to determine a player skill rating;
- xi. when to determine a player skill rating;
- xii. a type of benefit to provide in association with a player skill rating;
- xiii. an amount of a benefit to provide in association with a player skill rating; and/or
- xiv. any determination disclosed herein;

is/are predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a generated symbol or symbol combination, determined independent of a generated symbol or symbol combination, determined based on a random determination by the central controller, determined independent of a random determination by the central controller, determined based on a random determination at the gaming system, determined independent of a random determination at the gaming system, determined based on at least one play of at least one game, determined independent of at least one play of at least one game, determined based on a player's selection, determined independent of a player's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the player's primary game wager, determined independent of the player's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the player (i.e., a player tracking status), determined independent of a status of the player (i.e., a player tracking status), determined based on one or more other determinations disclosed herein, determined independent of any other determination disclosed herein or determined based on any other suitable method or criteria.

Gaming Systems

The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A "gaming system" as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices, such as desktop computers, laptop

computers, tablet computers or computing devices, personal digital assistants, mobile phones, and other mobile computing devices.

Thus, in various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more central servers, central controllers, or remote hosts; (b) one or more personal gaming devices in combination with one or more central servers, central controllers, or remote hosts; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more central servers, central controllers, or remote hosts in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single central server, central controller, or remote host; and/or (j) a plurality of central servers, central controllers, or remote hosts in combination with one another.

For brevity and clarity and unless specifically stated otherwise, "EGM" as used herein represents one EGM or a plurality of EGMs, "personal gaming device" as used herein represents one personal gaming device or a plurality of personal gaming devices, and "central server, central controller, or remote host" as used herein represents one central server, central controller, or remote host or a plurality of central servers, central controllers, or remote hosts.

As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in combination with a central server, central controller, or remote host. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the central server, central controller, or remote host through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a central server, central controller, or remote host through a data network.

In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a central server, central controller, or remote host, the central server, central controller, or remote host is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the central server, central controller, or remote host. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the central server, central controller, or remote host is configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the central server, central controller, or remote host and the EGM (or personal gaming device). The at least one processor of the central server,

central controller, or remote host is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the central server, central controller, or remote host. One, more than one, or each of the functions of the central server, central controller, or remote host may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the central server, central controller, or remote host.

In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host. In such "thin client" embodiments, the central server, central controller, or remote host remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such "thick client" embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices) are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a central server, central controller, or remote host, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host in a thin client configuration.

In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the central server, central controller, or remote host. In one example, the EGMs (or personal gaming devices) and the central server, central

controller, or remote host are located in a gaming establishment or a portion of a gaming establishment.

In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the central server, central controller, or remote host. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the central server, central controller, or remote host is located; or (b) in a gaming establishment different from the gaming establishment in which the central server, central controller, or remote host is located. In another example, the central server, central controller, or remote host is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a central server, central controller, or remote host and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN, though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the central server, central controller, or remote host identifies a player before enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique player name and password combination assigned to the player. The central server, central controller, or remote host may, however, identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the Internet browser of the EGM (or

personal gaming device). Examples of implementations of Internet-based gaming are further described in U.S. Pat. No. 8,764,566, entitled "Internet Remote Game Server," and U.S. Pat. No. 8,147,334, entitled "Universal Game Server".

The central server, central controller, or remote host and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for players to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

EGM Components

FIG. 3 is a block diagram of an example EGM **1000** and FIGS. 4A and 4B include two different example EGMs **2000a** and **2000b**. The EGMs **1000**, **2000a**, and **2000b** are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs **1000**, **2000a**, and **2000b**. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device **2000c** of FIG. 4C) may include some or all of the below components.

In these embodiments, the EGM **1000** includes a master gaming controller **1012** configured to communicate with and to operate with a plurality of peripheral devices **1022**.

The master gaming controller **1012** includes at least one processor **1010**. The at least one processor **1010** is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface **1006** of the master gaming controller **1012**; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices **1022** (such as input/output devices); and/or (5) controlling the peripheral devices **1022**. In certain embodiments, one or more components of the master gaming controller **1012** (such as the at least one processor **1010**) reside within a housing of the EGM (described below), while in other embodiments at least one component of the master gaming controller **1012** resides outside of the housing of the EGM.

The master gaming controller **1012** also includes at least one memory device **1016**, which includes: (1) volatile memory (e.g., RAM **1009**, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other

suitable forms); (2) non-volatile memory **1019** (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs **1008**); (4) read-only memory; and/or (5) a secondary memory storage device **1015**, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM disclosed herein. In certain embodiments, the at least one memory device **1016** resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device **1016** resides outside of the housing of the EGM. In these embodiments, any combination of one or more computer readable media may be utilized. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

The at least one memory device **1016** is configured to store, for example: (1) configuration software **1014**, such as all the parameters and settings for a game playable on the EGM; (2) associations **1018** between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable the at least one processor **1010** to communicate with the peripheral devices **1022**; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hipervlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller **1012** communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as periph-

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erals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller **1012** include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

As will be appreciated by one skilled in the art, aspects of the present disclosure may be illustrated and described herein in any of a number of patentable classes or context including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, aspects of the present disclosure may be implemented entirely hardware, entirely software (including firmware, resident software, microcode, etc.) or combining software and hardware implementation that may all generally be referred to herein as a “circuit,” “module,” “component,” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable media having computer readable program code embodied thereon.

Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the “C” programming language, Visual Basic, Fortran 2003, Perl, COBOL 2002, PHP, ABAP, dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the player’s computer, partly on the player’s computer, as a stand-alone software package, partly on the player’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the player’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other

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programmable instruction execution apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

In certain embodiments, the at least one memory device **1016** is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device **1016** of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, payable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In certain such embodiments, an operator (such as a gaming establishment operator) and/or a player uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

The at least one memory device **1016** also stores a plurality of device drivers **1042**. Examples of different types of device drivers include device drivers for EGM components and device drivers for the peripheral components **1022**. Typically, the device drivers **1042** utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet 175, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different type of card reader when device drivers for both card readers are stored in the at least one memory device.

In certain embodiments, the software units stored in the at least one memory device **1016** can be upgraded as needed. For instance, when the at least one memory device **1016** is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device **1016** from the master game controller **1012** or from some other external device. As another example, when the at least one memory device **1016** includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device **1016** can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another

example, when the at least one memory device **1016** uses flash memory **1019** or EPROM **1008** units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

In some embodiments, the at least one memory device **1016** also stores authentication and/or validation components **1044** configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, player input device components, information received from one or more player input devices, information stored in the at least one memory device **1016**, etc. Examples of various authentication and/or validation components are described in U.S. Pat. No. 6,620,047, entitled "Electronic Gaming Apparatus Having Authentication Data Sets".

In certain embodiments, the peripheral devices **1022** include several device interfaces, such as: (1) at least one output device **1020** including at least one display device **1035**; (2) at least one input device **1030** (which may include contact and/or non-contact interfaces); (3) at least one transponder **1054**; (4) at least one wireless communication component **1056**; (5) at least one wired/wireless power distribution component **1058**; (6) at least one sensor **1060**; (7) at least one data preservation component **1062**; (8) at least one motion/gesture analysis and interpretation component **1064**; (9) at least one motion detection component **1066**; (10) at least one portable power source **1068**; (11) at least one geolocation module **1076**; (12) at least one player identification module **1077**; (13) at least one player/device tracking module **1078**; and (14) at least one information filtering module **1079**.

The at least one output device **1020** includes at least one display device **1035** configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a player's player tracking status (as described below); (c) a secondary or upper display device in addition to the central display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash, account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM **2000a** illustrated in FIG. 4A includes a central display device **2116**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**. The example EGM **2000b** illustrated in FIG. 4B includes a central display device **2116**, an upper display device **2118**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**.

In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a display based

on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touch-screen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

In various embodiments, the at least one output device **1020** includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the player. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a ticket printer and dispenser **2136**. Examples of ticket-in ticket-out (TITO) technology are described in U.S. Pat. No. 5,429,361, entitled "Gaming Machine Information, Communication and Display System"; U.S. Pat. No. 5,470,079, entitled "Gaming Machine Accounting and Monitoring System"; U.S. Pat. No. 5,265,874, entitled "Cashless Gaming Apparatus and Method"; U.S. Pat. No. 6,729,957, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,729,958, entitled "Gaming System with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,736,725, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 7,275,991, entitled "Slot Machine with Ticket-In/Ticket-Out Capability"; and U.S. Pat. No. 6,048,269, entitled "Coinless Slot Machine System and Method".

In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the player following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the player in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the player; via a transfer of funds onto an electronically recordable identification card or smart card of the player; or via sending a virtual ticket having a monetary value to an electronic device of the player. Examples of providing payment using

virtual tickets are described in U.S. Pat. No. 8,613,659, entitled "Virtual Ticket-In and Ticket-Out on a Gaming Machine".

While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

In certain embodiments, the at least one output device **1020** is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a plurality of speakers **2150**. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the EGM. The videos may be customized to provide any appropriate information.

The at least one input device **1030** may include any suitable device that enables an input signal to be produced and received by the at least one processor **1010** of the EGM.

In one embodiment, the at least one input device **1030** includes a payment device configured to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a player identification card reader into which a player identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a combined bill and ticket acceptor **2128** and a coin slot **2126**.

In one embodiment, the at least one input device **1030** includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a player, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that player to fund the EGM. Examples of funding an EGM via communication between the EGM and a mobile device (such as a mobile phone) of a player are described in U.S. Patent Application Publication No. 2013/0344942, entitled "Avatar as Security Measure for Mobile Device Use with Electronic Gaming Machine". When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

In certain embodiments, the at least one input device **1030** includes at least one wagering or betting device. In various

embodiments, the one or more wagering or betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously-placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

In various embodiments, the at least one input device **1030** includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a player appropriately funds the EGM and places a wager, the EGM activates the game play activation device to enable the player to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a game play activation device in the form of a game play initiation button **2132**. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

In other embodiments, the at least one input device **1030** includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the EGM receives an actuation of the cashout device from a player and the player has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the player's credit balance. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a cashout device in the form of a cashout button **2134**.

In various embodiments, the at least one input device **1030** includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). The example EGMs **2000a**

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and **2000b** illustrated in FIGS. 4A and 4B each include a plurality of such buttons **2130**.

In certain embodiments, the at least one input device **1030** includes a touch-screen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touch-screen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

In embodiments including a player tracking system, as further described below, the at least one input device **1030** includes a card reader in communication with the at least one processor of the EGM. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a card reader **2138**. The card reader is configured to read a player identification card inserted into the card reader.

The at least one wireless communication component **1056** includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component **1056** transmits electrical, electromagnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

The at least one wired/wireless power distribution component **1058** includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component **1058** includes a magnetic induction system that is configured to provide wireless power to one or more player input devices near the EGM. In one embodiment, a player input device docking region is provided, and includes a power distribution component that is configured to recharge a player input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component **1058** is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

In certain embodiments, the at least one sensor **1060** includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor **1060** may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a predetermined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., players, casino employees, etc.), devices (e.g., player input devices), and/or systems within a predetermined proximity to the EGM.

The at least one data preservation component **1062** is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation system **1062** may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

The at least one motion/gesture analysis and interpretation component **1064** is configured to analyze and/or interpret information relating to detected player movements and/or gestures to determine appropriate player input information relating to the detected player movements and/or gestures.

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For example, in one embodiment, the at least one motion/gesture analysis and interpretation component **1064** is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a player; interpret the player's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the player; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

The at least one portable power source **1068** enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM **300** includes one or more rechargeable batteries.

The at least one geolocation module **1076** is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module **1076** is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module **1076** is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

The at least one player identification module **1077** is configured to determine the identity of the current player or current owner of the EGM. For example, in one embodiment, the current player is required to perform a login process at the EGM in order to access one or more features. Alternatively, the EGM is configured to automatically determine the identity of the current player based on one or more external signals, such as an RFID tag or badge worn by the current player and that provides a wireless signal to the EGM that is used to determine the identity of the current player. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized players from accessing confidential or sensitive information.

The at least one information filtering module **1079** is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays **1035** of the EGM.

In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, solenoids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. U.S. Pat. No. 7,290,072 describes a variety of EGMs including one or more communication ports that enable the EGMs to communicate and operate with one or more external peripherals.

As generally described above, in certain embodiments, such as the example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM. Further, the EGM is configured such that a player may operate it while standing or sitting. In various embodiments, the EGM is

positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a player may operate typically while sitting. As illustrated by the different example EGMs 2000a and 2000b shown in FIGS. 4A and 4B, EGMs may have varying housing and display configurations.

In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained approval from a regulatory gaming commission.

The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

Operation of Primary or Base Games and/or Secondary or Bonus Games

In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as “primary games”) and/or any secondary or bonus games or other functions (referred to herein as “secondary games”) displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a player; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a player.

As generally explained above, in various embodiments in which the gaming system includes a central server, central controller, or remote host and a changeable EGM, the at least one memory device of the central server, central controller, or remote host stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one changeable EGM as a secondary game to be played simultaneously with a play of a primary game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

In operation of such embodiments, the central server, central controller, or remote host is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of

the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the central server, central controller, or remote host to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award. Examples of this type of award evaluation are described in U.S. Pat. No. 7,470,183, entitled “Finite Pool Gaming Method and Apparatus”; U.S. Pat. No. 7,563,163, entitled “Gaming Device Including Outcome Pools for Providing Game Outcomes”; U.S. Pat. No. 7,833,092, entitled “Method and System for Compensating for Player Choice in a Game of Chance”; U.S. Pat. No. 8,070,579, entitled “Bingo System with Downloadable Common Patterns”; and U.S. Pat. No. 8,398,472, entitled “Central Determination Poker Game”.

In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided,

the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards. Examples of this type of award determination are described in U.S. Pat. No. 7,753,774, entitled "Using Multiple Bingo Cards to Represent Multiple Slot Paylines and Other Class III Game Options"; U.S. Pat. No. 7,731,581, entitled "Multi-Player Bingo Game with Multiple Alternative Outcome Displays"; U.S. Pat. No. 7,955,170, entitled "Providing Non-Bingo Outcomes for a Bingo Game"; U.S. Pat. No. 8,070,579, entitled "Bingo System with Downloadable Common Patterns"; and U.S. Pat. No. 8,500,538, entitled "Bingo Gaming System and Method for Providing Multiple Outcomes from Single Bingo Pattern".

In certain embodiments in which the gaming system includes a central server, central controller, or remote host and an EGM, the EGM is configured to communicate with the central server, central controller, or remote host for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the central server, central controller, or remote host monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the central server, central controller, or remote host. In this embodiment, the accounting and gaming information system includes: (a) a player database configured to store player profiles, (b) a player tracking module configured to track players (as described below), and (c) a credit system configured to provide automated transactions. Examples of such accounting systems are described in U.S. Pat. No. 6,913,534, entitled "Gaming Machine Having a Lottery Game and Capability for Integration with Gaming Device Accounting System and Player Tracking System," and U.S. Pat. No. 8,597,116, entitled "Virtual Player Tracking and Related Services".

As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) may comprise any suitable games and/or wagering games, such as, but not limited to: electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system. In certain such

embodiments, the gaming system includes one or more paylines associated with the reels. The example EGM 2000b shown in FIG. 4B includes a payline 1152 and a plurality of reels 1154. In certain embodiments, one or more of the reels are independent reels or unisymbol reels. In such embodiments, each independent reel generates and displays one symbol.

In various embodiments, one or more of the paylines is horizontal, vertical, circular, diagonal, angled, or any suitable combination thereof. In other embodiments, each of one or more of the paylines is associated with a plurality of adjacent symbol display areas on a requisite number of adjacent reels. In one such embodiment, one or more paylines are formed between at least two symbol display areas that are adjacent to each other by either sharing a common side or sharing a common corner (i.e., such paylines are connected paylines). The gaming system enables a wager to be placed on one or more of such paylines to activate such paylines. In other embodiments in which one or more paylines are formed between at least two adjacent symbol display areas, the gaming system enables a wager to be placed on a plurality of symbol display areas, which activates those symbol display areas.

In various embodiments, the gaming system provides one or more awards after a spin of the reels when specified types and/or configurations of the indicia or symbols on the reels occur on an active payline or otherwise occur in a winning pattern, occur on the requisite number of adjacent reels, and/or occur in a scatter pay arrangement.

In certain embodiments, the gaming system employs a ways to win award determination. In these embodiments, any outcome to be provided is determined based on a number of associated symbols that are generated in active symbol display areas on the requisite number of adjacent reels (i.e., not on paylines passing through any displayed winning symbol combinations). If a winning symbol combination is generated on the reels, one award for that occurrence of the generated winning symbol combination is provided. Examples of ways to win award determinations are described in U.S. Pat. No. 8,012,011, entitled "Gaming Device and Method Having Independent Reels and Multiple Ways of Winning"; U.S. Pat. No. 8,241,104, entitled "Gaming Device and Method Having Designated Rules for Determining Ways To Win"; and U.S. Pat. No. 8,430,739, entitled "Gaming System and Method Having Wager Dependent Different Symbol Evaluations".

In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs, the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award. Examples of progressive gaming systems are described in U.S. Pat. No. 7,585,223, entitled "Server Based Gaming System Having Multiple Progressive Awards"; U.S. Pat. No. 7,651,392, entitled "Gaming Device System Having Partial Progressive Payout"; U.S. Pat. No. 7,666,093, entitled "Gaming Method and Device Involving Progressive Wagers"; U.S. Pat. No. 7,780,523, entitled "Server Based Gaming System Having Multiple Progressive Awards"; and U.S. Pat. No. 8,337,298, entitled "Gaming Device Having Multiple Different Types of Progressive Awards".

As generally noted above, in addition to providing winning credits or other awards for one or more plays of the primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of player excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a "BONUS" symbol appearing on three adjacent reels along a payline following a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, that is obtained, a given number of secondary game wagering points or credits is accumulated in a "secondary game meter" configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple "buy-in." For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager "buys-in" to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game or a wager of a designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable players of those EGMs to work in conjunction with one another, such as by enabling the players to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable players of those EGMs to compete against one another for one or more awards. In one such embodiment, the EGMs enable the players of those EGMs to participate in one or more gaming tournaments for one or more awards. Examples of group gaming systems are described in U.S. Pat. No. 8,070,583, entitled "Server Based Gaming System and Method for Selectively Providing One or More Different Tournaments"; U.S. Pat. No. 8,500,548, entitled "Gaming System and Method for Providing Team Progressive Awards"; and U.S. Pat. No. 8,562,423, entitled "Method and Apparatus for Rewarding Multiple Game Players for a Single Win".

In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a player's gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a player is issued a player identification card that has an encoded player identification number that uniquely identifies the player. When the player's playing tracking card is inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the player identification number off the player tracking card to identify the player. The gaming system timely tracks any suitable information or data relating to the identified player's gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different

embodiments, for one or more players, the player tracking system includes the player's account number, the player's card number, the player's first name, the player's surname, the player's preferred name, the player's player tracking ranking, any promotion status associated with the player's player tracking card, the player's address, the player's birthday, the player's anniversary, the player's recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device. Examples of player tracking systems are described in U.S. Pat. No. 6,722,985, entitled "Universal Player Tracking System"; U.S. Pat. No. 6,908,387, entitled "Player Tracking Communication Mechanisms in a Gaming Machine"; U.S. Pat. No. 7,311,605, entitled "Player Tracking Assembly for Complete Patron Tracking for Both Gaming and Non-Gaming Casino Activity"; U.S. Pat. No. 7,611,411, entitled "Player Tracking Instruments Having Multiple Communication Modes"; U.S. Pat. No. 7,617,151, entitled "Alternative Player Tracking Techniques"; and U.S. Pat. No. 8,057,298, entitled "Virtual Player Tracking and Related Services".

Web-Based Gaming

In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the player must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an "app") installed on the personal gaming device before the player can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices (such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers, the one or more servers make various determinations based on the inputs and determine content to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

In certain such embodiments, the one or more servers must identify the player before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the player must identify herself to the one or more servers, such as by inputting the player's unique playername and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

Once identified, the one or more servers enable the player to establish an account balance from which the player can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the player to initiate an electronic funds transfer to transfer funds from a

bank account to the player's account balance. In other embodiments, the one or more servers enable the player to make a payment using the player's credit card, debit card, or other suitable device to add money to the player's account balance. In other embodiments, the one or more servers enable the player to add money to the player's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the player to cash out the player's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the player, or by initiating printing of a voucher at a kiosk in a gaming establishment.

In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out players' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the player's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

If the payment server determines that the player's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the player that the player's account balance is too low to place the desired wager. If the payment server determines that the player's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server. The game server then determines an outcome and any associated award for the play of the game. The game server notifies the payment server of any nonzero award, and the payment server increases the player's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area.

In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device. Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the player to

play games on the EGM remotely via the personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area. Examples of tethering an EGM to a personal gaming device and geo-fencing are described in U.S. Patent Appl. Pub. No. 2013/0267324, entitled "Remote Gaming Method Allowing Temporary Inactivation Without Terminating Playing Session Due to Game Inactivity".

Social Network Integration

In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a player's gaming experience with the player's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the player's wall, newsfeed, or similar area of the social networking website accessible by the player's connections (and in certain cases the public) such that the player's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the player's likes or dislikes or the player's list of connections. In certain embodiments, the gaming system enables the player to link the player's player account to the player's social networking account(s). This enables the gaming system to, once it identifies the player and initiates a gaming session (such as via the player logging in to a website (or an application) on the player's personal gaming device or via the player inserting the player's player tracking card into an EGM), link that gaming session to the player's social networking account(s). In other embodiments, the gaming system enables the player to link the player's social networking account(s) to individual gaming sessions when desired by providing the required login information.

For instance, in one embodiment, if a player wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to play). In another embodiment, if a player joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network sever to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to fill the vacancy). In another embodiment, if the player consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see. In another embodiment, the gaming system enables the player to recommend a game to the player's connections

by posting a recommendation to the player's wall (or other suitable area) of the social networking website.

Differentiating Certain Gaming Systems from General Purpose Computing Devices

Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

For instance, EGMs are highly regulated to ensure fairness and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are implemented in EGMs that differ significantly from those of general purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the player, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the player. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general purpose computing device.

A second difference between EGMs and general purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has

been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a player of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

A third difference between EGMs and general purpose computing devices is authentication—EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code. Examples of EGM code authentication are described in U.S. Pat. No. 6,962,530, entitled "Authentication in a Secure Computerized Gaming System"; U.S. Pat. No. 7,043,641, entitled "Encryption in a Secure Computerized Gaming System"; U.S. Pat. No. 7,201,662, entitled "Method and Apparatus for Software Authentication"; and U.S. Pat. No. 8,627,097, entitled "System and Method Enabling Parallel Processing of Hash Functions Using Authentication Checkpoint Hashes".

A fourth difference between EGMs and general purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software com-

ponents and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally-operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to "re-trigger" the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In other words, the watchdog timer always functions from the time power is applied to the board.

Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply, but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

As described above, certain EGMs are state-based machines. Different functions of the game provided by the EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the player's wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or all do

not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write access characteristics (such as at least 5.08 Mbytes/sec (Read) and/or at least 38.0 Mbytes/sec (Write)). Memory devices that meet or exceed the above criteria may be referred to as "fault-tolerant" memory devices.

Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly, battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a player is required to make a number of selections on a video display screen. When a malfunction has occurred after the player has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the player. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of

the graphical presentation that was previously presented on the EGM and the state of the EGM (e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a player may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the player was correct or not in the player's assertion. Examples of a state-based EGM, recovery from malfunctions, and game history are described in U.S. Pat. No. 6,804,763, entitled "High Performance Battery Backed RAM Interface"; U.S. Pat. No. 6,863,608, entitled "Frame Capture of Actual Game Play"; U.S. Pat. No. 7,111,141, entitled "Dynamic NV-RAM"; and U.S. Pat. No. 7,384,339, entitled, "Frame Capture of Actual Game Play".

Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that differ from the "standard" EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The pur-

pose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third party verification device. Once the trusted memory device is verified as authentic, and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives. Examples of trusted memory devices are described in U.S. Pat. No. 6,685,567, entitled "Process Verification".

In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., "unalterable memory") such as EPROMS, PROMS, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the identity of the trusted information source. For example, the trusted information source and the remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities.

EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information might be designed to detect tampering attempts and clear or erase itself when an attempt at tampering has been detected. Examples of trusted memory devices/sources are described in U.S. Pat. No. 7,515,718, entitled "Secured Virtual Network in a Gaming Environment".

Mass storage devices used in a general purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present. Examples of using a mass storage device are described in U.S. Pat. No. 6,149,522, entitled "Method of Authenticating Game Data Sets in an Electronic Casino Gaming System".

Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without

departing from the spirit and scope of the present subject matter and without diminishing its intended technical scope. It is therefore intended that such changes and modifications be covered by the appended claims.

The invention is claimed as follows:

1. A method of operating a gaming establishment player rating system, the method comprising:

responsive to an occurrence of a triggering event at a first point in time when a player identified via a player tracking identification number electronically assigned to that player is associated with a first player skill rating based on a first skill score that is derived from a quantifiable skill input made at an independently operating gaming establishment device prior to the first point in time and that is independent of any individual outcome displayed in association with any play of any skill-based game, displaying, by a display device, a first resource to allocate to the identified player, the first resource being based on the first player skill rating, and responsive to an occurrence of the triggering event at a second, subsequent point in time when the identified player is associated with a second, higher player skill rating based on a second, higher skill score, displaying, by the display device, a second resource to allocate to the identified player, wherein the second resource is based on the second, higher player skill rating, and a value of the second resource is lower than a value of the first resource.

2. The method of claim 1, wherein the first point in time occurs responsive to a conclusion of a session.

3. The method of claim 1, wherein the second, higher skill score is derived from another quantifiable skill input made at another independently operating gaming establishment device after the first point in time and is independent of any individual outcome displayed in association with any play of any skill-based game.

4. The method of claim 3, wherein the second, higher skill score is based on a comparison of the other quantifiable skill input to an optimal skill input.

5. The method of claim 1, wherein the first resource is based on a first range of resources having a first average expected value that is independent of any play of any skill-based game.

6. The method of claim 5, wherein the second resource is based on a second range of resources having a second average expected value that is independent of any play of any skill-based game, the second average expected value being lower than the first average expected value.

7. A method of operating a gaming establishment player rating system, the method comprising:

responsive to a player identified via a player tracking identification number electronically assigned to that player being associated with a first player skill rating based on a first skill score, independent of any play of any game, displaying, by a display device, a first level of reinvestment resources to be allocated to the identified player, the first level of reinvestment resources being based on the first player skill rating, and

after a modification of the first player skill rating associated with the identified player to a second, higher player skill rating, independent of any play of any game, displaying, by the display device, a second, lower level of reinvestment resources to be allocated to the identified player, the second, lower level of reinvestment resources being based on the second, higher player skill rating.

8. The method of claim 7, wherein an allocation of the second, lower level of reinvestment resources to the identified player automatically occurs responsive to the modification of the first player skill rating associated with the identified player to the second, higher player skill rating. 5

9. The method of claim 7, wherein an allocation of the second, lower level of reinvestment resources to the identified player occurs responsive to an input made by a gaming establishment operator.

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